ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH

TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN

KHOA MẠNG MÁY TÍNH VÀ TRUYỀN THÔNG

Computer Networking

Lab 03: TCP/UDP Protocol

A. Overview

1. Goals

- Learn how to stream a video over the network with VLC and learn the behavior of TCP - UDP protocol.
- Use Wireshark to capture packets and analyze TCP and UDP packet characteristics over streaming media protocol.
- Learn how TCP uses sequence numbers and acknowledgment numbers are used for reliable data transmission.

2. Knowledge base

- Knowledge of TCP - UDP protocol of Transport chapter.

3. Experimental environment & tools

- An Internet-connected computer using the Windows/Linux operating system.
- **Wireshark** software: *Students can download the latest version for free by following the instructions at: https://www.wireshark.org/download.html*

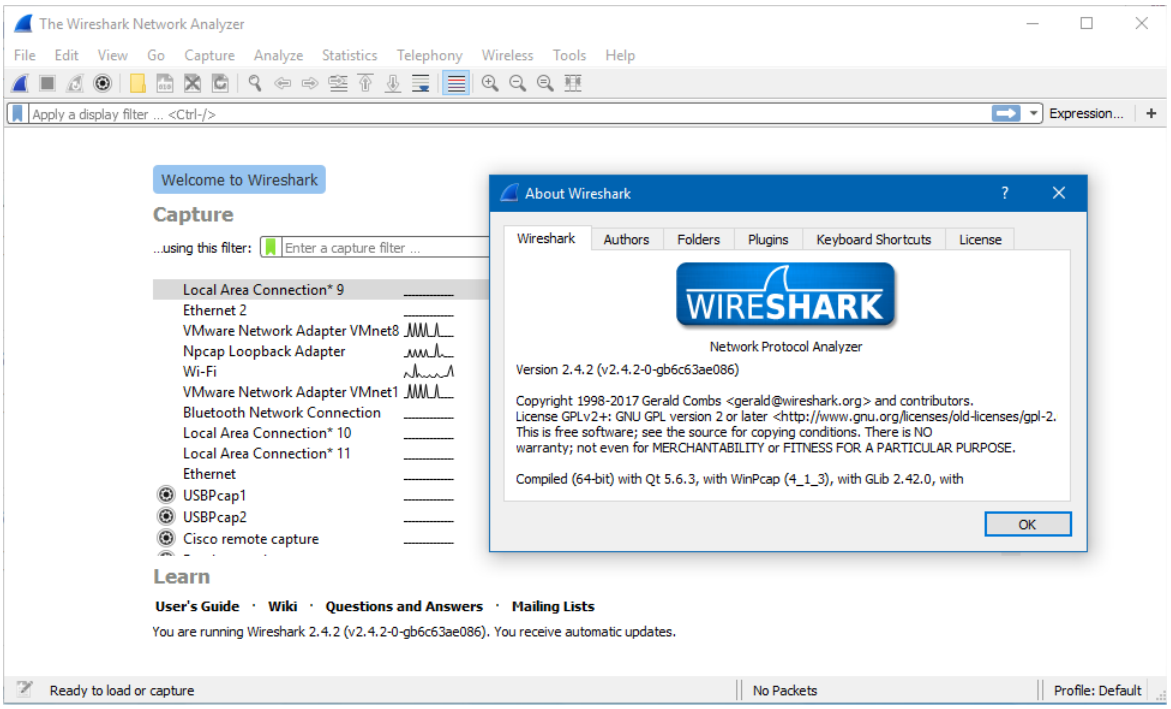

Fig 1. Main UI of Wireshark.

- **VLC media player** software: *Students can download the new version for free at https://www.videolan.org/vlc/index.html*

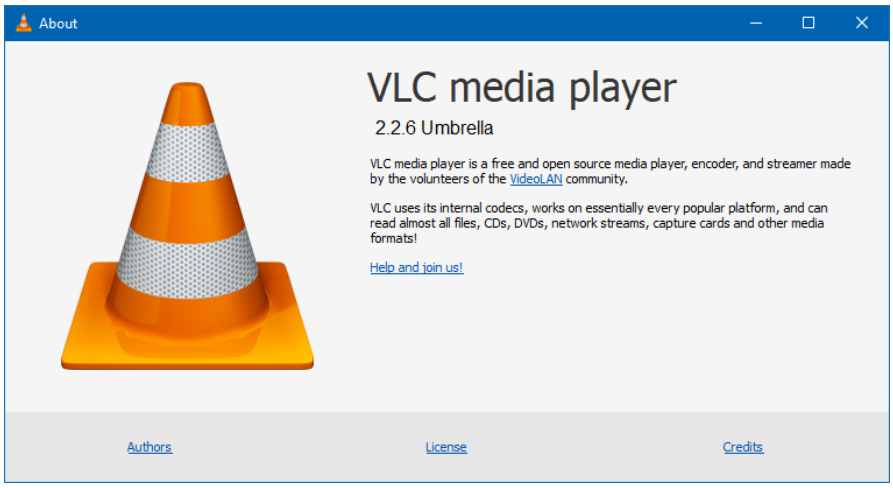

Fig 2. VLC Media Player.

B. Practise

1. Task 1: General Analysis of TCP and UDP Protocols

1.1 Analysis and Packet Capture During Streaming Video (Attempt 1)

- **Step 1:** Open Wireshark and commence packet capture.
- **Step 2:** Access a video streamed from a server by opening the VLC media player

  » Select **Open network stream**.

  Enter the address in the format: **rtsp://[** server IP address**]:8554/**

  Example when the server's IP is **192.168.1.100**:
  *The server's IP address will be provided in class.

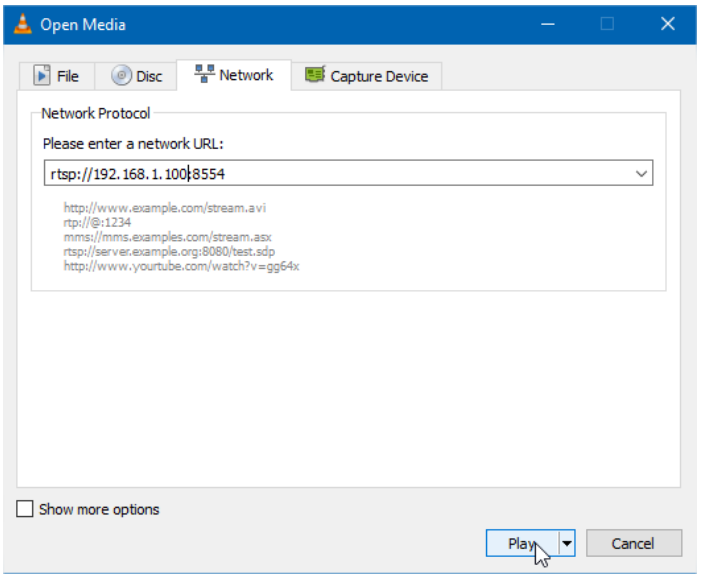

*Fig 3. Select **Play** to begin viewing the video being streamed from the Server.*

- **Step 3:** Observe the displayed video.
- **Step 4:** Terminate the packet capture in Wireshark.

1.2 Analysis of Results from Attempt 1

Q1: Describe the observed results when streaming the video from the Server?

Q2: Based on the captured Wireshark file, does the streaming process in this scenario utilize the UDP or TCP protocol?

1.3 Analysis and Packet Capture During Streaming Video (Attempt 2)

- **Step 5:** Open Wireshark and commence packet capture.
- **Step 6:** Access a video streamed from a server by opening the VLC media player

  » Select **Open network stream**.

  Enter the address in the format: **http://[**server IP address**]:8080/**
- **Step 7:** Observe the displayed video.
- **Step 8:** Terminate the packet capture in Wireshark.

1.4 Analysis of Results from Attempt 2

Q3: Describe the observed results when streaming the video from the Server?

Q4: Based on the captured Wireshark file, does the streaming process in this scenario utilize the UDP or TCP protocol?

Q5: Analyze the difference between the two video streaming attempts.

2. Task 2: Analysis of UDP Protocol Operation

2.1 Perform a DNS Query and Capture UDP Packets

DNS is an application layer protocol that uses both **UDP** and **TCP** at the Transport Layer. **DNS** stands for Domain Name System, meaning the system for resolving domain names. Simply put, DNS is fundamentally a system that converts website domain names, such as *www.tenmien.com,* into a corresponding numerical IP address, and vice versa.

➲ *Students record the IP configuration information of the PC:*

Use the command **ipconfig /all** in the Command Prompt interface to find and record the following information in the table below:

Tab 1. IP configuration information of the PC Example

IPv4 Link-local IPv6 address	192.168.1.8 fe80::4923:18cd:8c3f:afbe
MAC address	14-DD-A9-BF-48-AB
Default gateway	fe80::1 192.168.1.1
DNS Servers	fe80::1 8.8.8.8

- **Địa chỉ MAC** (Physical Address) and **IP** (IPv4 và Link-local IPv6 Address) of the Network Interface Card (NIC) used by the student for network communication.
- IP addresses of the specified default gateways.
- IP addresses of the DNS servers specified for the PC.

Q6: Students are to find and fill in the table according to the template above.

➲ *Perform Capture of DNS Query and Response Packets:*

- **Step 1:** Open Command Prompt.
- **Step 2:** Ensure DNS Cache is cleared by typing the command **ipconfig /flushdns**
- **Step 3:** Open Wireshark and commence packet capture.
- **Step 4:** From the Command Line, type **nslookup -type=A uit.edu.vn** (or another chosen domain).
- **Step 5:** After the result is displayed, stop the Wireshark capture and save the file with the name format MSSV-**UDP.pcapng** *(to be submitted with the report).*

1. Through guidance, research, and present the fundamental differences, advantages, and disadvantages of UDP and TCP. Through observing the video

streaming, research and present the answers to Q1, 2, 3, 4, 5. Find and complete Q6 to be used as information, contributing to a clearer understanding of DNS and TCP content below.

2.2 Analysis of UDP Protocol Operation

Open the Wireshark file captured in Task 2 above, filter for "*dns*" packets, and answer the following questions:

2. In the list of captured packets, locate the packet querying the domain **`uit.edu.vn`** (or the domain chosen in Task 2, step 4 above).

   Hint: Contains "*standard query*" and "*A uit.edu.vn*".

3. Identify the response packet to the above query? From the response message, record the **IP address** of the domain **uit.edu.vn**
4. Select one UDP packet, identify the *fields* present in the UDP header, and explain the significance of each field? **Hint**: Refer to the User Datagram Protocol section.
5. Based on the information displayed in Wireshark, determine the length (*in bytes*) of each field in the UDP header?
6. What does the value of the **Length** field in the UDP header represent the length of? Provide evidence to support this claim.
7. What is the maximum number of bytes that the UDP **payload** (*the section containing the original data, excluding the UDP header and IP header*) can contain?

   Hint: Based on the size of the Length field in the UDP header and the maximum value it can represent.

8. What is the maximum possible value for the **Source port**?

9. Observe the two packets found in **2.** and **3.**, and describe the relationship between the IP addresses and ports of these two packets.

   Hint: Observe the **Source (IP, Port)** and **Destination (IP, Port)** of the two packets.

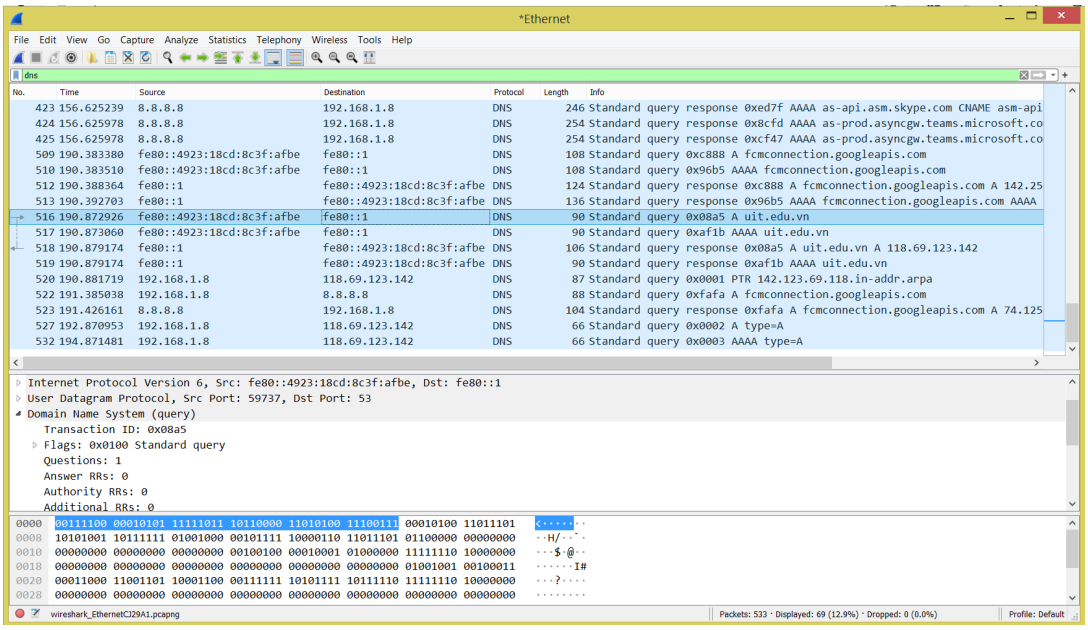

Fig 4. UDP Analysis Example

3. Task 3: Analysis of TCP Protocol Operation

3.1 Accessing a Website and Capturing TCP Packets

- **Step 1:** Clear the cache of the browser being used.
- **Step 2:** Open Wireshark and commence packet capture.
- **Step 3:** Using a browser, access the following address:

  http://www.celuit.edu.vn/
- **Step 4:** Once the website is fully displayed, stop the Wireshark capture and save the file with the name format MSSV-**TCP.pcapng**. *(to be submitted with the report).*

3.2 Analysis of TCP Protocol Operation

Open the Wireshark file captured in Task 3 above, filter for "tcp" packets, and answer the following questions:

10. Find the IP address and TCP port of the Client machine?
11. Find the IP address of the Server. Which port does it use to receive **segments**?
12. Locate the **TCP SYN segment** (TCP packet with the SYN flag set) that initiates the TCP connection between the client and the server. Which component in the segment indicates that it is a **TCP SYN segment**?
    Hint: Observe the Flags field.
13. Find the **sequence number** of the **SYN/ACK segment** sent by the server to the client in response to the SYN segment.

    Find the value of the **Acknowledgement** in the SYN/ACK segment.

    How can the server determine this value? Which component in the segment indicates that it is a SYN/ACK segment?
14. Identify the first 6 segments that the server sends to the Client (*based on the Packet Number – No*)
    - Find the sequence number of those first 6 segments.
    - Determine the time when each segment was sent and the time when the ACK for each segment was received.
    - Calculate the difference between the time each segment was sent and the time the ACK for each segment was received by calculating the **RTT (Round Trip Time)** for these 6 segments.

Round-trip time (RTT) is the time interval from when the computer begins sending a segment until it receives the corresponding ACK. See more in the Transport Layer Slides.

Note: Create a statistical table and calculate the time values in the following format*:*

No.	Time	RTT	SEQ number	ACK number	Classification
					Sent Packet / Received ACK

** Requires clear evidence with images, identified by the collected packets.*

15. Was any segment retransmitted? What information during the transmission process indicates this? Provide an explanation.

Gợi ý: *To check the amount of data transmitted in a unit of time, instead of manually calculating from the packet data, use a Wireshark feature – Statistics » TCP Stream Graph » Time-Sequence-Graph (Stevens).*

Select any segment in the packet list section.

The resulting graph will be similar to the example:

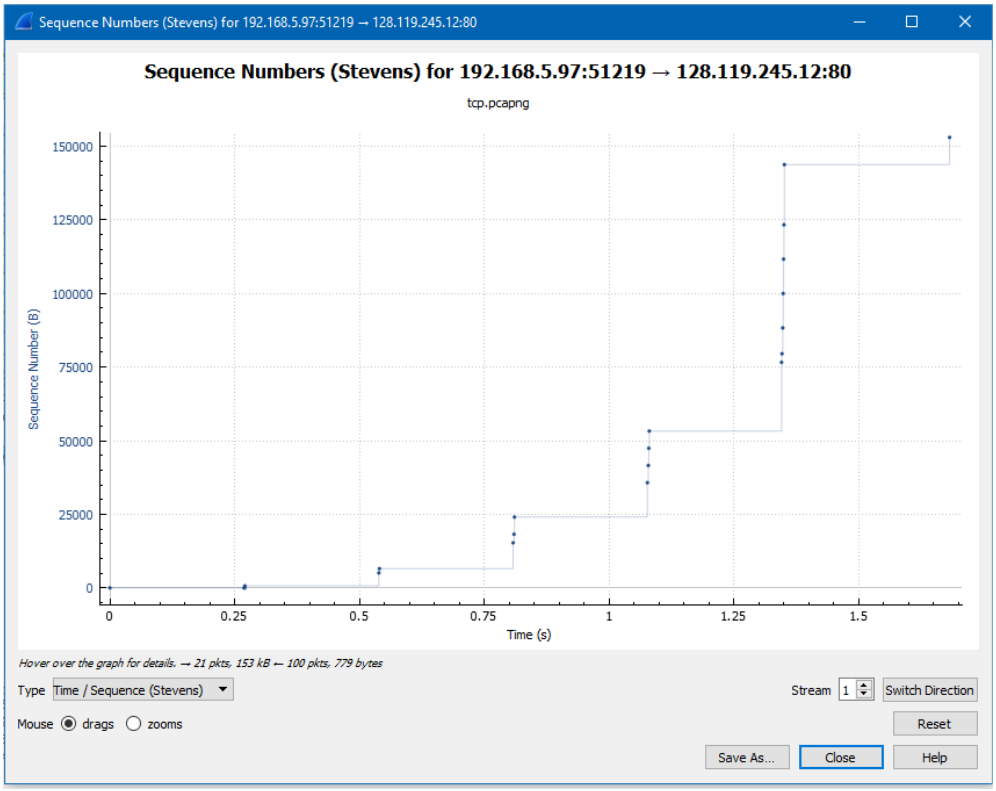

Fig 5. Ví dụ về biểu đồ Sequence Number (Stevens)

Each dot on the graph represents a TCP segment with its sequence number corresponding to the time the segment was sent. Note that a cluster of dots corresponds to a sequence of packets sent consecutively.

THE END